Challenge Task: Q&A Model Development

Weightage: 5% (i.e., Half of the total lab evaluation (10% weightage distribution))

This is a group assignment. You may form your own team of up to 4 members, or consider the members of the currently assigned project group. You need to fill out the group members in this form: https://forms.gle/FntUyMYKUuRkJSPB9

Your objective is to build a Question-and-Answer (Q&A) model that effectively answers queries using the provided training dataset. You can design your own architecture, and it is encouraged.

Dataset Overview:
You will be given a Q&A training set (X, Y) (Dataset link: QnA):

- X represents the query text passed as input to your Q&A model.
- Y contains the desired responses corresponding to the Q&A model.

In addition to the Q&A dataset, you are provided with the sources of the responses and a collection of crop image samples. These images serve as optional contextual information, and you are encouraged to use them creatively to enhance your model's responses to the queries.

You are required to implement your Q&A model and evaluate it on the training dataset (split into 90% train and 10% test sets) using appropriate metrics, comparing its performance against at least one baseline model. Prepare a report on the assessment of your model.

Evaluation Criteria:
A new set of queries (X_t) will be shared with you for the evaluation of your 2 models on unseen data. Your model's responses will be evaluated using our chosen evaluation metrics that will not be disclosed in advance. Based on the evaluation score, your group will be ranked.

- Groups in the top 5 will be awarded a full mark 100%.
- Groups in the top 6 - 10 will be awarded 80% of the total mark
- Groups in the top 11 - 15 will be awarded 70% of the total mark
- Groups in the top 16 - 25 will be awarded 60% of the total mark
- Groups in the top 26 - 35 will be awarded 50% of the total mark
- The rest of the groups that have submitted the requirements will be awarded 40% of the total mark

This assignment aims to enhance your understanding of NLP through the development and evaluation of a Q&A model in a practical context.

Files to be submitted:

- Your Kaggle or Collab notebook codes
- Evaluation report of the given training dataset
- Updated CSV of the test queries (X_t) with their corresponding responses.
- A separate notebook to load your final trained checkpoints and generate responses for test queries (X_t) as well as your test set.
- **A major discrepancy in the reported results will result in a ZERO grade.**

Good luck!